

Practice Problems Set-2

Problem 1

Let

$$f(x, y) = e^{xy} + \ln(1 + x^2 + y^2),$$

where

$$x = r \cos \theta, \quad y = r \sin \theta.$$

1. Compute $\frac{\partial f}{\partial r}$ and $\frac{\partial f}{\partial \theta}$ using the chain rule.

2. Verify that

$$\nabla_{(r,\theta)} f = J^T \nabla_{(x,y)} f,$$

where J is the Jacobian matrix of (x, y) with respect to (r, θ) .

3. Evaluate the gradient at $(r, \theta) = (1, \pi/4)$.

Problem 2

Let

$$F(x, y) = \begin{pmatrix} x^2 y + e^{xy} \\ \sin(xy) + y^3 \end{pmatrix}.$$

1. Compute the Jacobian matrix $DF(x, y)$.
2. Compute the directional derivative of F at $(1, -1)$ in the direction $v = (2, 1)$.
3. Determine whether F is locally invertible at $(0, 1)$.

Problem 3

Let

$$f(x, y) = \begin{cases} \frac{x^3 y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

1. Compute $f_x(0, 0)$ and $f_y(0, 0)$.
2. Compute $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$.
3. Determine whether $f_{xy}(0, 0) = f_{yx}(0, 0)$.

Problem 4

Let

$$f(x, y) = e^{x+y} \cos(xy).$$

1. Compute all first and second order partial derivatives at $(0, 0)$.
2. Find the second order Taylor polynomial of f at $(0, 0)$.
3. Approximate $f(0.1, -0.1)$ using the polynomial.

Problem 5

Let

$$g(u, v) = \ln(1 + u^2 + v^2),$$

where

$$u = x + y, \quad v = x - y.$$

1. Compute ∇g in terms of u, v .
2. Use the chain rule to compute $\nabla_{(x,y)} g(x, y)$.
3. Compute the second order Taylor expansion of $g(x, y)$ at $(0, 0)$.
4. Approximate $g(0.05, 0.02)$.